

Preliminary Requirements

1 Required Skills

In this course, you will make **extensive use of the command line (terminal)** and Docker containers.

For the first exercise session, please ensure that you have a working Docker setup (see Section 3) and a good understanding of the following:

- Basic usage of the Bash command line interface in Linux. This is important, the **Docker containers do not have a graphical user interface!**
- Concept and basic usage of Git (version control system).
- Optional: Basic usage of SSH (how to connect to a remote machine).

If you are not familiar with these topics, there are several free courses and documentation to catch up on the basics.

- Bash: <https://www.youtube.com/playlist?list=PLAwTw4SYaPlBwWBVoIWKfSWxuhfSW-D>
- Git: <https://www.udacity.com/course/version-control-with-git--ud123>
- SSH: <https://www.fim.uni-passau.de/en/it-services/remote-access/remote-login-via-ssh/>

2 FIM Account

In the lab sessions, we will use a git repository hosted by the Faculty of Computer Science and Mathematics (FIM). To access this repository you need a FIM account, which is different from your standard university account (the ZIM account). To create a FIM account, follow the instructions on [this site](https://www.fim.uni-passau.de/en/it-services/login-and-account/fim-accounts/)¹.

3 Docker Installation

To install Docker on your machine, follow the instructions on <https://docs.docker.com/get-docker/>. Make sure to run the hello-world image as described in the Docker documentation to verify that your Docker environment is installed correctly.

¹<https://www.fim.uni-passau.de/en/it-services/login-and-account/fim-accounts/>

3.1 Docker Server

If you plan to use the lab computers or your own device does not support Docker, you may use our *Docker server*. **If you choose to install and run Docker on your own device, you can ignore this subsection.**

To request access to the Docker server, please email your **FIM account** name to sds-tutorials@uni-passau.de with the subject “Docker Server Access” by **April 21, 2026**.

After server access is granted, follow the steps on “Docker Server Instructions”.

4 Questions?

Use the course forum in Stud.IP for questions and technical problems. Before posting a new question, check whether your question or problem has already been answered or solved.

When asking for help with technical problems, please provide enough relevant details, such as your operating system, the failed command, etc.

Note: As every machine has a different environment (operating system, applications installed, etc.) and different hardware, we can only provide limited support for technical problems on your own device.